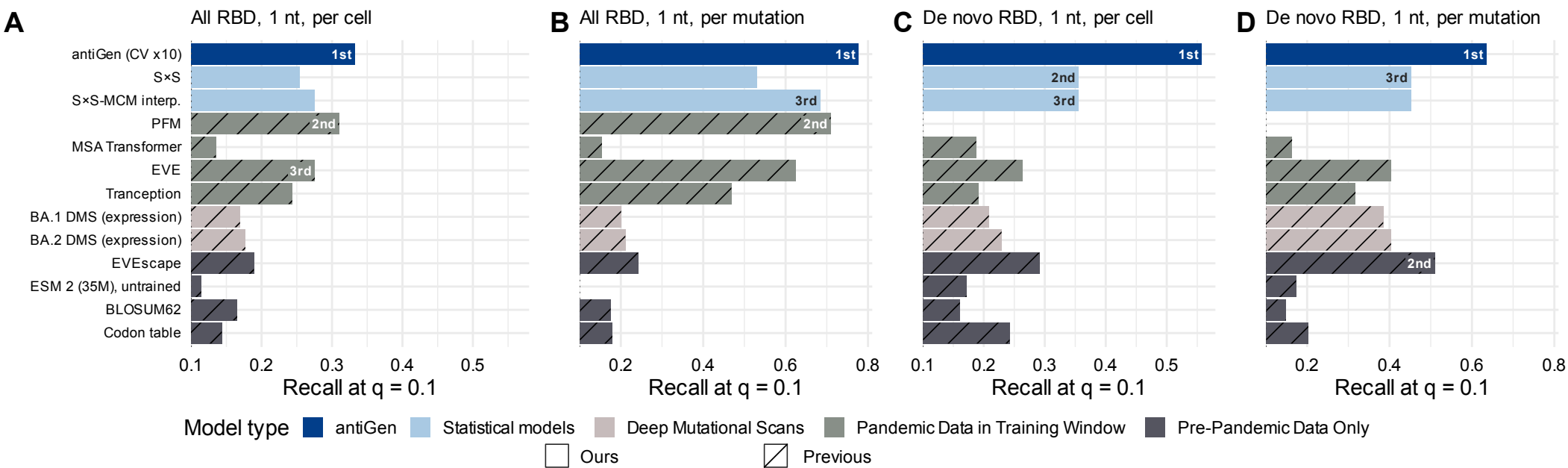
A
All RBD, 1 nt, per cell
B
All RBD, 1 nt, per mutation
C
De novo RBD, 1 nt, per cell
D
De novo RBD, 1 nt, per mutation
antiGen (CV x10)
S×S
S×S-MCM interp.
PFM
MSA Transformer
EVE
Tranception
BA.1 DMS (expression)
BA.2 DMS (expression)
EVEscape
ESM 2 (35M), untrained
BLOSUM62
Codon table
1st
2nd
3rd
0.1
0.2
0.3
0.4
0.5
0.6
0.8
Recall at q = 0.1
Model type
antiGen
Statistical models
Deep Mutational Scans
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ours
Previous